CUSTOMER SEGMENTATION VIA UNSUPERVISED CLUSTERING & DIMENSION REDUCTION

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ABSTRACT

Understanding customer behavior is essential for targeted marketing and personalized service delivery in modern retail environments. This project analyzed behavioral and demographic data from retail customers to identify different segments of their behavior using unsupervised machine learning and dimensionality reduction techniques. To preserve variance while removing redundant correlations among high- dimensional features, Principal Component Analysis (PCA) evaluated three distinct clustering approaches-K-Means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Hierarchical DBSCAN(HDBSCAN) to assess cluster stability, spatial density variations and noise handling. Model performance was evaluated using Silhouette Coefficients and Davies-Bouldin Index scores. The optimal K-Means model successfully identified distinct customer profiles, providing actionable segments for tailored marketing strategies.

PROJECT OVERVIEW AND RATIONALE

Retail businesses generate vast amounts of customer transaction and interaction data. Traditional rule-based segmentation often fails to capture subtle multi-dimensional patterns, leading to ineffective marketing campaigns or misallocated resources.

This project is designed to address three core analytical questions:

- How effectively can PCA compress high-dimensional behavioral features while critical variance is preserved?
- How do partitioning methods like K-Means compare against density-based algorithms like DBSCAN and HDBSCAN when identifying natural customer groupings?
- Which evaluation metrics offer the most reliable validation of cluster quality when ground-truth labels were absent?

EXPLORING AND PREPROCESSING THE DATASET

The dataset comprises of customer attributes capturing demographic details and spending patterns:

- Customer ID: Unique identifier per entry.
- Gender: Categorical variable indicating gender identity.
- Age: Continuous variable representing customer age.
- Annual Income(k$): Continuous metric reflecting yearly earnings in thousands of dollars.
- Spending score(1-100): Score assigned by the retail platform based on customer behavior and their respective purchasing pattern.

PREPROCESSING PIPELINE

- Data integrity and cleaning: Checked for missing values and duplicates; no incomplete rows were detected.
- Feature transformation: Encoded categorical attributes into numeric vectors.
- Feature Scaling: Applied StandardScaler (Z-score standardization) across numerical attributes to ensure equal weighting across distance metrics, preventing features with larger numerical scales from dominating spatial distances.
- Dimensionality Reduction: Performed PCA to map correlated multi-dimensional inputs into orthogonal principal components capturing over 80% of total variance within the first two components for simplified dimensionality reduction.

CLUSTERING METHODOLOGY AND EXPERIMENTAL RESULTS

I evaluated three distinct unsupervised clustering algorithms to determine the optimal grouping structure:

Method 1: K-Means Clustering

- Implemented the Elbow Method (Sum of square errors) and silhouette analysis across a range of k values (k= 2 to 10) to identify the optimal cluster count.
- Optimal k=5 yielded well-separated clusters with minimum intra-cluster variance.
- Silhouette Score: 0.554
- Davies-Bouldin Index: 0.578

Method 2: DBSCAN (Density-Based Spatial Clustering)

- Tested varying neighborhood radii(eps) and minimum sample thresholds (min_samples) to identify arbitrary-shaped dense regions while isolating noise/outliers.
- . Identified core dense clusters but classified peripheral customers with lower density as noise points.
- Silhouette Score: 0.382.

Method 3: HDBSCAN (Hierarchical DBSCAN)

- Applied hierarchical density estimation to adaptively discover clusters across varying density gradients without requiring a single global distance threshold.
- Successfully isolated noise and identified stable core density hierarchies, though boundaries remained soft for intermediate income earners.

KEY CUSTOMER SEGMENTS IDENTIFIED

The 5- cluster K-Means solution revealed clear, actionable customer personas:

- High Income, High Spending ("Target/Premium Customers"): High annual earnings with high spending scores: ideal for premium products and loyalty programs.

- High Income, Low Spending (“Careful/ High Potential”): Substantial income but conservative spending; prime targets for value driven promotional engagement.
- Low Income, High spending (“Careless/ Impulse Buyers”): Lower income levels with high spending scores; responsive to frequent discounts and trends.
- Low Income, Low Spending (“Sensible/ Budget-Conscious”): Lower income and low spending scores; price-sensitive and focused on essential purchases.
- Average Income, Average Spending (“ Standard/ Core Majority”): Moderate earnings and moderate spending; form the baseline volume of the customer pool.

CONCLUSIONS AND PRACTICAL APPLICATIONS

- Dimensionality reduction enhances geometry: PCA effectively eliminated feature redundancy and collinearity, improving the spatial separation of spatial algorithms.
- Algorithm suitability depends on data geometry: While DBSCAN excelled at density filtering and noise detection, K-Means provided clearer, more interpretable partitions for structured retail data where hyper-spherical clusters naturally align with business segments.
- Business Value: These segments allow marketing teams to tailor campaign strategies, optimize promotional budgets, and design personalized customer retention programs based on empirical behavioral data.